Gravity above the surface

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use distance from a planet's center to compare fields at altitude.

SUCCESS CRITERIA - I can...

1. Use r=R+h rather than altitude alone.
2. Compare fields with the inverse-square relation.
3. Explain why orbit is not a zero-gravity region.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two fixed masses attract with **75 N** at separation r. **Find the force** at separation **5** r and **explain the factor**.

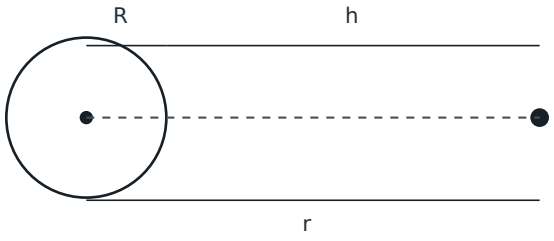
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $r = R + h$; $g_h/g_{surface} = [R/(R + h)]^2$.
R is planet radius; h is altitude. g_h is field strength at altitude h.

Worked reasoning: At altitude h=**2** R above a planet's surface, center distance is r=R+h=**3** R. A surface field of **18 N/kg** becomes **18/9=2 N/kg**.

Your turn: Draw the radius and altitude as different line segments. **Mark the full center-to-object distance**.

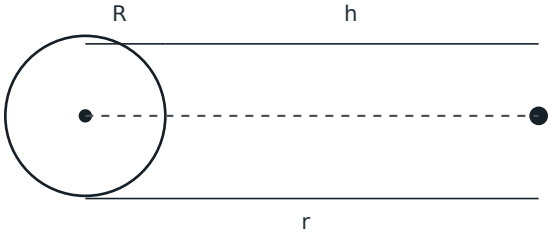


Measure r from the center: r = R + h.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A satellite is one planet-radius above the surface.
1. **Label R**, h and r on a sketch.
2. What is r in units of R?

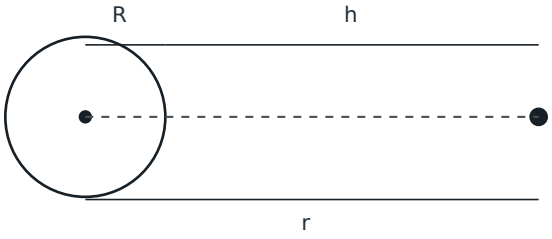


Label surface radius R, altitude h and center distance r.

Marzano check - circle: ① ② ③

04 Apply the physics

Surface field is **12 N/kg**. At height h=R, center distance doubles. **Calculate the new field**: **12÷4**.



Marzano check - circle: ① ② ③

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05 Explain your reasoning

Explain why an astronaut at modest orbital altitude can float even though gravity is only slightly weaker there.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Maximum static friction is **60 N**. A **24 N** push leaves a box still. **Find actual friction**. A **72 N** push starts sliding with **48 N** kinetic friction; **find net force**.
Draw the forces on the chosen object and **label their sources**.



Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

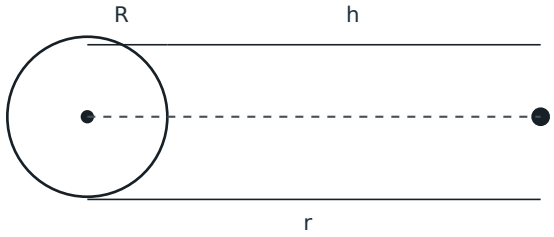
A planet has surface g=**9 N/kg**. At center distance **3** R, what isg?

Field / surface field	r / R	h / R
___	___	___
___	___	___

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A planet's field is **16 N/kg** at its surface. At what altitude in planet radii is it **4 N/kg**? **Use a sketch**.



Marzano check - circle: 1 2 3